Multi-Agent Systems and Agentic AI
Practical Examination Report

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Repository and logs. The code used for Part I is available at GitLab repository. Training and evaluation logs are available at W&B run. The adaptive-planning implementation studied in Part II is linked at cmbagent_lg adaptive-planning branch. Replace these placeholders with exact clickable URLs, including the exact commit where required.

Notes on jointly owned work. Sections I.1–I.3 describe experiments run using the shared team TPU and shared repository. The analysis and prose in this report are my own. Section I.4 and Part II are completed individually.

Contents

Part I: GRPO finetuning of Gemma 3 on a TPU

I.1 Reproducing the baseline

Table 1: Baseline run details.

Item	Value
Repository commit	`TODO`
TPU type	v6e-1
Training command	`python scripts/train.py`
Evaluation command	`python scripts/evaluate.py`
Wall-clock time	[TODO: e.g. 6 h 20 min]
GRPO steps completed	[TODO: number of steps]
Held-out split	[TODO: e.g. GSM8K test split]
Evaluation seed	[TODO: fixed seed]

Run details.

Baseline patches. [TODO: State whether the baseline ran as shipped. If not, document the minimal fix required, with the affected file and commit. Keep this short.]

Table 2: Baseline GSM8K accuracy.

Model	Split	Seed	Accuracy
Base model, `google/gemma-3-1b-it`	[TODO: split]	[TODO: seed]	[TODO: value]
LoRA GRPO checkpoint	[TODO: split]	[TODO: seed]	[TODO: value]

figures/baseline_reward_kl.pdf

Figure 1: Baseline training curves showing mean reward $\bar{r}$ and $\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$ against GRPO step.

Evaluation.

I.2 Understanding the baseline

Policies in the LoRA setup. [TODO: Explain what plays the role of π_θ, π_{old} and π_{ref}, and identify which carries trainable parameters.]

Group size and advantage estimator. [TODO: State the group size K from `scripts/config.py`. Identify the Tunix advantage estimator and state whether it is standard group-mean or leave-one-out.]

Reward terms. [TODO: Identify shaping rewards and the true correctness reward in `scripts/rewards.py`. Explain what would happen early in training with correctness reward alone in terms of within-group reward variance σ_r.]

Clipped surrogate. [TODO: Locate where the PPO-style clipped surrogate is implemented in Tunix. State what ε corresponds to in `scripts/config.py` and note any deviation from the lecture form, such as per-token rather than per-sequence ratios.]

I.3 Improving on the baseline

Modification and motivation. [TODO: Describe the principled modification. Examples: leave-one-out advantage, sweep K, change β or ε, modify shaping rewards, add length penalty, change LoRA rank, learning rate, schedule or generation length. Connect the choice to GRPO theory.]

Controlled comparison. [TODO: State how the comparison was controlled: same data, same evaluation split, same total compute or normalised per step. Mention baseline plus at least two further complete runs, or justify fewer.]

Table 3: Controlled comparison on held-out GSM8K. Include uncertainty using a second seed or a bootstrap confidence interval over evaluation prompts.

Model / run	Modification	Seed(s)	Accuracy	Uncertainty
Base model	None	[TODO:]	[TODO:]	[TODO:]
Baseline GRPO	Default config	[TODO:]	[TODO:]	[TODO:]
Variant 1	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Variant 2	[TODO:]	[TODO:]	[TODO:]	[TODO:]

`figures/reward_kl_variants.pdf`

Figure 2: Training curves for the baseline and variants on shared axes, showing mean reward and KL over GRPO step.

figures/grpo_diagnostic.pdf

Figure 3: Diagnostic plot for a known GRPO failure mode, such as advantage distribution, degenerate reward groups, response length, entropy, or KL versus accuracy.

Discussion. [TODO: Connect the empirical findings to lecture theory. State whether the observed behaviour matched the prediction, and diagnose any negative results honestly.]

I.4 GRPO theory

Question 1

(i) Since $\mathbb{E}[\bar{r}] = \mu$,

$$\mathbb{E}[\hat{A}_i] = \frac{\mathbb{E}[r_i] - \mathbb{E}[\bar{r}]}{\sigma} = 0, \qquad \mathbb{E}[X_i] = h_i \mathbb{E}[\hat{A}_i] = 0, \qquad \mathbb{E}[\hat{g}] = \frac{1}{K}\sum_{i=1}^{K} \mathbb{E}[X_i] = 0.$$

This expectation averages only reward noise around fixed completions. In GRPO/RLHF the completions are sampled from the policy, so the score $\nabla_\theta \log \pi_\theta(a \mid q)$ can be correlated with the reward of the sampled completion and the policy gradient need not vanish.

(ii) The common random term is the group mean $\bar{r}$, which appears in every advantage:

$$X_i = h_i \hat{A}_i = \frac{h_i}{\sigma}(r_i - \bar{r}).$$

For $i \neq j$,

$$\operatorname{Cov}(X_i, X_j) = \frac{h_i h_j^\top}{\sigma^2} \operatorname{Cov}(r_i - \bar{r}, r_j - \bar{r}),$$
$$\operatorname{Cov}(r_i - \bar{r}, r_j - \bar{r}) = 0 - \frac{\sigma^2}{K} - \frac{\sigma^2}{K} + \frac{\sigma^2}{K} = -\frac{\sigma^2}{K}.$$

Therefore

$$\boxed{\operatorname{Cov}(X_i, X_j) = -\frac{1}{K} h_i h_j^\top, \qquad i \neq j.}$$

The negative sign is the price of using a group-centred contrast: if one completion is above the group mean, the others are pushed below that same mean. This removes prompt-level reward offsets in GRPO, but the K rollouts are no longer K independent gradient samples.

(iii) Since $\operatorname{Var}(X_i) = (1 - K^{-1}) h_i h_i^\top$ and the off-diagonal covariances are as above, with $\bar{h} = K^{-1} \sum_i h_i$,

$$\operatorname{Var}(\hat{g}) = \frac{1}{K^2} \left[\sum_i h_i h_i^\top - \frac{1}{K} \left(\sum_i h_i \right) \left(\sum_j h_j \right)^\top \right] = \frac{1}{K^2} \sum_i (h_i - \bar{h})(h_i - \bar{h})^\top. \tag{1}$$

The naive i.i.d. calculation would miss the subtraction of $K^{-1} (\sum_i h_i)(\sum_j h_j)^\top$. To turn the covariance comparison into a scalar effective sample size, fix a direction u and define

$$K_{\text{eff}}(u) := \frac{u^\top \operatorname{Var}(X_1) u}{u^\top \operatorname{Var}(\hat{g}) u} = \frac{K(K-1)(u^\top h_1)^2}{\sum_i \{u^\top (h_i - \bar{h})\}^2}, \tag{2}$$

with $K_{\text{eff}}(u) = \infty$ when the denominator vanishes. A rotation-invariant trace summary is

$$K_{\text{eff,tr}} := \frac{\operatorname{tr} \operatorname{Var}(X_1)}{\operatorname{tr} \operatorname{Var}(\hat{g})} = \frac{K(K-1) \|h_1\|^2}{\sum_i \|h_i - \bar{h}\|^2}.$$

For comparable non-degenerate scores, the denominator grows as $O(K)$, so $K_{\text{eff}} = O(K)$, but the constant is determined by diversity of the score vectors, not just by rollout count. If $h_i = c_i v$ are all aligned, **??** is rank one and only scalar variation in the c_i is learnable from the group contrast. If all h_i are identical then $\hat{g} = h_1 K^{-1} \sum_i \hat{A}_i = 0$ exactly: group normalisation has removed all fixed-prompt signal.

Question 2

(i) Assume $\pi_t(a) > 0$ on its support and, in the measurable case, that the normaliser below is finite. Any optimiser must have support contained in that of π_t, since otherwise $\operatorname{KL}(\pi \| \pi_t) = \infty$. With λ enforcing $\sum_a \pi(a) = 1$, stationarity gives

$$A_t(a) - \frac{1}{\eta} \left(\log \frac{\pi(a)}{\pi_t(a)} + 1 \right) + \lambda = 0.$$

Normalising the resulting exponential tilt yields the unique optimiser

$$\pi_{t+1}(a) = \frac{\pi_t(a) \exp\{\eta A_t(a)\}}{Z_t}, \qquad Z_t = \sum_b \pi_t(b) \exp\{\eta A_t(b)\}. \tag{3}$$

The objective is strictly concave on this support, so the optimiser is unique. This is the KL mirror-descent or exponential-tilting update: small η gives a conservative RLHF step near π_t, while large η concentrates probability on high-advantage responses but cannot put mass on responses assigned zero probability by π_t.

(ii) Let $F_t(\pi) = \mathbb{E}_{a\sim\pi}[A^{\pi_t}(a)] - \eta^{-1}\,\mathrm{KL}(\pi\|\pi_t)$. The defining property of the true advantage is $\mathbb{E}_{a\sim\pi_t}[A^{\pi_t}(a)] = 0$, so $F_t(\pi_t) = 0$. Optimality of **??** then implies

$$J(\pi_{t+1}) = \mathbb{E}_{\pi_{t+1}}[A^{\pi_t}] \geq \frac{1}{\eta}\,\mathrm{KL}(\pi_{t+1}\|\pi_t) \geq 0 = J(\pi_t).$$

Thus the exact unrestricted update is monotone in this advantage objective. In neural-network GRPO the exponential-tilted distribution may be outside the model or LoRA-adapter family, and the advantage is a finite-sample, group-normalised estimate. Either approximation, especially with degenerate groups where σ_r is tiny or zero, can break the guarantee.

(iii) For $\hat{A}_t > 0$, clipping removes the incentive to increase the sampled ratio beyond $1+\varepsilon$; for $\hat{A}_t < 0$, it removes the incentive to decrease it below $1-\varepsilon$. The resulting trust region is advantage-dependent, samplewise, and soft: it clips the objective on sampled actions, but does not enforce the ratio bounds after a shared neural update. By contrast, $\mathrm{KL}(\pi_\theta(\cdot \mid q)\|\pi_{\mathrm{old}}(\cdot \mid q)) \leq \delta$ is one hard, distribution-level constraint; it can permit a large change on some actions provided the joint KL budget remains satisfied. In LLM RLHF this means PPO clipping can miss unsampled completions and prompt-level KL spikes, whereas a KL constraint directly budgets total distributional drift.

Question 3

(i) For any integrand f, $\mathbb{E}_{\pi_{\mathrm{old}}}[f] = \mathbb{E}_{\pi_{\mathrm{mix}}}[wf]$, with

$$w_i = \frac{\pi_{\mathrm{old}}(a_i \mid q)}{\pi_{\mathrm{mix}}(a_i \mid q)} = \frac{\pi_{\mathrm{old}}(a_i \mid q)}{(1-\alpha)\pi_{\mathrm{old}}(a_i \mid q) + \alpha\pi_{\mathrm{unif}}(a_i \mid q)}. \tag{4}$$

Thus the direct importance-corrected version of the stated, group-normalised estimator is

$$\hat{g}_{\mathrm{mix}} = \frac{1}{K}\sum_{i=1}^{K} w_i \nabla_\theta \log \pi_\theta(a_i \mid q)\,\frac{r_i - \bar{r}}{\sigma_r}, \qquad a_i \overset{\mathrm{iid}}{\sim} \pi_{\mathrm{mix}}. \tag{5}$$

(ii) For each fixed realised action, $\pi_{\mathrm{mix}}(a_i \mid q) \to \pi_{\mathrm{old}}(a_i \mid q)$ and hence $w_i \to 1$. The realised r_i, $\bar{r}$, and σ_r are unchanged, so **??** tends to $\hat{g}_{\mathrm{GRPO}}$.

(iii) Linearity of expectation gives

$$V^{\pi_{\mathrm{mix}}}(q) = (1-\alpha)V^{\pi_{\mathrm{old}}}(q) + \alpha V^{\pi_{\mathrm{unif}}}(q).$$

Consequently, multiplying by w_i outside the advantage does not change the fact that the unweighted $\bar{r}$ estimates $V^{\pi_{\mathrm{mix}}}$ rather than $V^{\pi_{\mathrm{old}}}$. For the fixed-scale target $\mathbb{E}_{\pi_{\mathrm{old}}}[\nabla_\theta \log \pi_\theta(a \mid q)(R(q,a) - V^{\pi_{\mathrm{old}}})/s]$, use the leave-one-out weighted baseline

$$\hat{V}_{-i} = \frac{1}{K-1}\sum_{j\neq i} w_j r_j, \qquad \hat{g}^{\mathrm{LOO}}_{\mathrm{mix}} = \frac{1}{K}\sum_{i} w_i \nabla_\theta \log \pi_\theta(a_i \mid q)\frac{r_i - \hat{V}_{-i}}{s}. \tag{6}$$

Indeed, $\mathbb{E}_{\pi_{\mathrm{mix}}}[\hat{V}_{-i}] = V^{\pi_{\mathrm{old}}}$ and $\hat{V}_{-i}$ is independent of a_i, so **??** is unbiased for that old-policy score–advantage expectation. The common self-normalised baseline $\hat{V}_{\mathrm{SN}} = \sum_j w_j r_j / \sum_j w_j$ is often lower variance and consistent, but it is biased at finite K; using the same group's empirical σ_r adds the usual GRPO normalisation bias and fails when the group reward variance is zero.

(iv) There are two variance effects. First, the unweighted reward variance under the proposal is

$$\begin{aligned}\mathrm{Var}_{\pi_{\mathrm{mix}}}(R) = (1-\alpha)\,\mathrm{Var}_{\pi_{\mathrm{old}}}(R) + \alpha\,\mathrm{Var}_{\pi_{\mathrm{unif}}}(R)\\ + \alpha(1-\alpha)\big(V^{\pi_{\mathrm{old}}} - V^{\pi_{\mathrm{unif}}}\big)^2.\end{aligned}$$

Thus mixing increases advantage variance when the uniform sampler has much larger reward variance or a very different mean, as with rare correct answers mixed with many invalid LLM completions. It decreases raw advantage variance when $V^{\pi_{\text{unif}}} \simeq V^{\pi_{\text{old}}}$ and $\mathrm{Var}_{\pi_{\text{unif}}}(R) < \mathrm{Var}_{\pi_{\text{old}}}(R)$, for example if uniform exploration smooths a spiky old-policy reward distribution.

Second, importance weighting changes the gradient estimator variance. For a scalar projection Y of the score–advantage term,

$$\mathbb{E}_{\pi_{\text{mix}}}[w^2 Y^2] = \mathbb{E}_{\pi_{\text{old}}}[w Y^2].$$

Therefore lower raw reward variance does not guarantee lower gradient variance: large-$|Y|$ samples in regions where $\pi_{\text{old}} > \pi_{\text{mix}}$ get $w > 1$ and can dominate, while putting proposal mass on high-signal responses that π_{old} under-samples can reduce variance by making their weights smaller.

(v) Under the stated assumption,

$$w_i = c^T = \exp(T \log c) \longrightarrow 0 \quad \text{exponentially as } T \to \infty.$$

When $\alpha > 0$, mixture samples include tokens to which the uniform component assigns more probability than π_{old}; more formally, $\mathbb{E}_{\pi_{\text{mix}}}[\log(\pi_{\text{old}}/\pi_{\text{mix}})] = -\mathrm{KL}(\pi_{\text{mix}} \| \pi_{\text{old}}) < 0$, so the geometrically typical token ratio is below one. Consequently most long responses have negligible weight while a few rare trajectories dominate, which is disastrous for long-form LLM RLHF because the effective sample size collapses exponentially with response length. One concrete fix is to scale the mixture coefficient as $\alpha = \gamma/T$, keeping the cumulative proposal perturbation, and hence typical log-weight magnitude, $O(1)$; the trade-off is weaker uniform exploration for longer responses.

Part II: Adaptive planning in `cmbagent_lg`

II.1 Workflow diagram

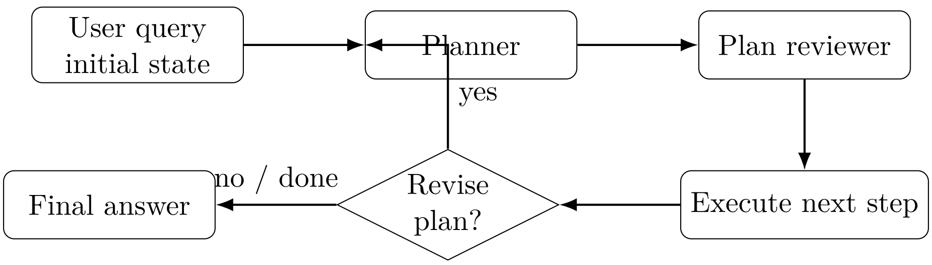

Figure 4: Placeholder adaptive-planning workflow. Replace this with the exact LangGraph nodes, edges, conditional routing, and state fields from the studied commit. This diagram does not count towards the 2-page Part II limit.

Walkthrough. [TODO: In at most half a page, explain what triggers a plan revision, what part of the plan may change, what is held fixed across a revision, and how the loop terminates. Include the exact commit link studied.]

II.2 Problems

Problem 1: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

Problem 2: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

Problem 3: [TODO: short name]. [TODO: Argue the limitation or failure mode, grounded in planning and closed-loop control theory.]

II.3 Proposed improvements

Improvement 1. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

Improvement 2. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

Improvement 3. [TODO: State what changes, why it addresses the corresponding problem, and what it trades off.]

References

- Hu, E. J. et al. (2021). *LoRA: Low-Rank Adaptation of Large Language Models*. `https://arxiv.org/abs/2106.09685`
- Tunix documentation or source link used for GRPO implementation: `https://github.com/google/tunix`
- Exact `cmbagent_lg` adaptive-planning commit studied: [TODO: insert clickable commit URL]

A Appendix

A.1 Additional run details

[TODO: Optional non-essential logs, extra tables, hyperparameter dumps, and supplementary figures.]